

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Problem Statement #19 | Healthcare & Telemedicine

Personalized Meal & Diet Subscription Manager

Target Stakeholders / Actors: Subscriber, Dietitian

1. Requirements Table

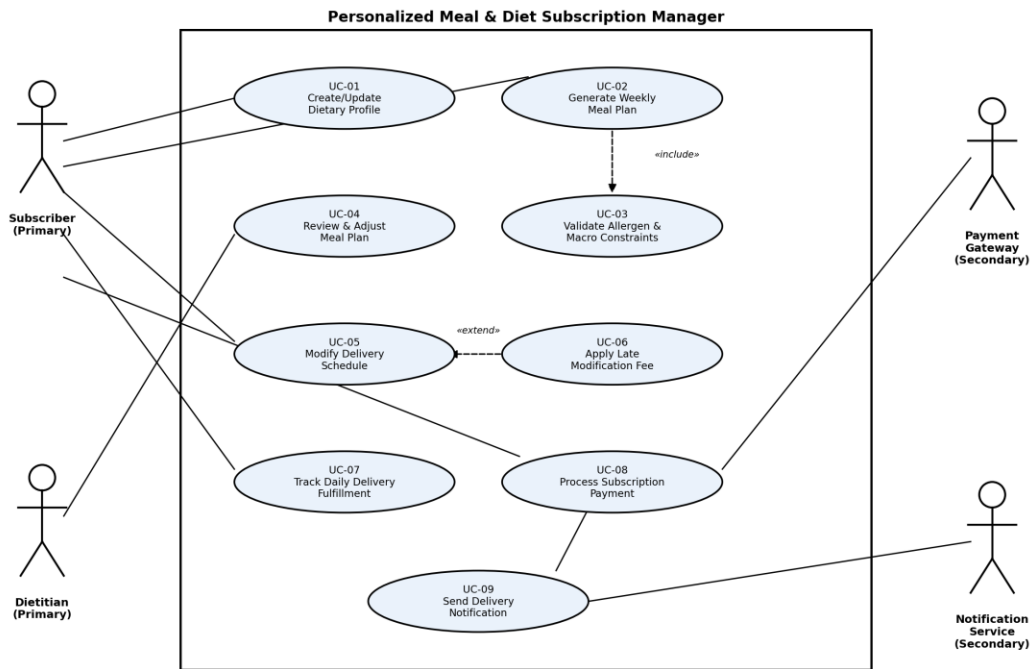
Five Functional Requirements (FR-001–FR-005) and two Nonfunctional Requirements (NFR-001–NFR-002), each specified with Req ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate weekly meal plans strictly adhering to user macro-nutrient constraints and allergen exclusion lists.	High	Pass: Generated menu contains zero excluded allergen ingredients. Fail: Peanuts included for a peanut-allergic profile.	Core patient-safety requirement; prevents allergic reactions and non-compliant diets.
FR-002	Functional	The system shall allow the Dietitian to review and approve or reject each subscriber's generated weekly meal plan before it is finalized.	High	Pass: 100% of finalized meal plans carry a Dietitian approval ID and timestamp. Fail: A menu is dispatched without an approval record.	Provides clinical oversight to confirm menus are medically appropriate.
FR-003	Functional	The system shall allow a subscriber to pause, skip, or reschedule an upcoming delivery from their account.	High	Pass: A submitted pause/skip/reschedule request updates the delivery calendar within 5 seconds and shows as "Modified" in order history. Fail: Requested change is not reflected.	Gives subscribers flexibility and reduces wasted deliveries.
FR-004	Functional	The system shall display real-time daily delivery fulfillment status (Preparing, Dispatched, Delivered) to the subscriber.	Medium	Pass: Status updates to "Delivered" within 15 minutes of courier confirmation. Fail: Status remains "Dispatched" more than 15 minutes after confirmed delivery.	Builds subscriber trust and reduces support inquiries.
FR-005	Functional	The system shall automatically notify the subscriber and the Dietitian when a scheduled meal plan cannot be fulfilled due to an allergen conflict or ingredient stock-out.	High	Pass: Both parties are notified within 2 minutes of the conflict being detected. Fail: No notification is generated for a detected conflict.	Prevents unsafe substitutions and keeps clinical staff informed.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional	The subscription engine must support schedule modifications (pause, skip, reschedule) up to 12 hours before dispatch.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures the system stays responsive and secure during high-demand periods while protecting patient health data.
NFR-002	Nonfunctional	The system shall maintain at least 99.5% monthly uptime for the meal-plan generation and ordering service.	High	Pass: Monthly uptime monitoring logs show $\geq$ 99.5% availability. Fail: Logged downtime exceeds 0.5% of the month.	Subscribers depend on timely weekly menus; extended outages directly disrupt medically prescribed diets.

2. UML Use-Case Diagram

The diagram below models the system boundary, primary and secondary actors, all use cases, and the required «include» and «extend» relationships (created with the same modelling approach as draw.io / Lucidchart).



Actors

- Subscriber (Primary) – end user managing their dietary profile, subscription, and deliveries.
- Dietitian (Primary) – clinical staff who review and approve generated meal plans.
- Payment Gateway (Secondary) – external system that processes subscription payments.
- Notification Service (Secondary) – external system that delivers alerts to the subscriber.

Use Cases

- UC-01: Create/Update Dietary Profile
- UC-02: Generate Weekly Meal Plan
- UC-03: Validate Allergen & Macro Constraints — «include» of UC-02 (always executed when a plan is generated)
- UC-04: Review & Adjust Meal Plan (Dietitian)
- UC-05: Modify Delivery Schedule (pause/skip/reschedule)
- UC-06: Apply Late Modification Fee — «extend» of UC-05 (only when a change is requested inside the 12-hour cutoff)
- UC-07: Track Daily Delivery Fulfillment
- UC-08: Process Subscription Payment
- UC-09: Send Delivery Notification

3. Use-Case Flow Specification

Use Case: Generate Weekly Meal Plan

Preconditions:

- The Subscriber has an active subscription.
- The Subscriber's dietary profile (allergies, calorie goals, diabetic limits) has been completed and verified.

Main Success Scenario:

1. System retrieves the Subscriber's metabolic profile and dietary constraints.
2. System generates a candidate weekly menu.
3. System validates the menu against the allergen exclusion list and macro-nutrient constraints («include» UC-03: Validate Allergen & Macro Constraints).
4. System routes the candidate meal plan to the Dietitian's review queue.
5. Dietitian reviews and approves the meal plan.
6. System finalizes and publishes the weekly menu to the Subscriber.
7. System schedules the daily deliveries associated with the approved menu.
8. Subscriber receives a notification that the meal plan is ready.
9. Use case ends successfully.

Alternate Flow:

3a. Allergen or Macro Constraint Violation

- 3a1. System detects an excluded allergen or a macro-nutrient constraint violation in the candidate menu.
- 3a2. System discards the candidate menu and automatically regenerates a new one.
- 3a3. If a violation persists after 3 regeneration attempts, the system flags the profile for manual Dietitian review.

5a. Dietitian Rejects Meal Plan

- 5a1. Dietitian rejects the plan and records comments.
- 5a2. System regenerates the menu incorporating the Dietitian's comments.
- 5a3. Revised plan is resubmitted to the Dietitian for review (returns to step 5).

Postconditions:

- An approved weekly meal plan is stored and associated with the Subscriber's account.
- A delivery schedule has been created for the approved menu.
- The Subscriber has been notified that the meal plan is ready.